

# Abhijit Chowdhary

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## About Me

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I consider myself a computational mathematician and my research interests lie in wielding math to design algorithms for real-world problems. I practice that in developing scalable algorithms to quantify, analyze, and control the uncertainty inherent in mathematical models. During my Ph.D., I focused on the uncertainty quantification (UQ) of large-scale inverse problems for physical phenomena governed by partial differential equations (PDEs). At the moment, I'm pursuing new directions in computational math, via problem structure-exploiting neural architectures, uncertainty quantification, randomized algorithms, among more.

## Education

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- North Carolina State University**, PhD in Applied Mathematics – Raleigh, NC Aug 2020 – May 2025
- Recipient of the Siewert Fellowship
- New York University**, BA in Mathematics, Computer Science – New York, NY May 2016 – May 2020
- Minor in Classics

## Experience

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- Postdoctoral Visiting Scholar**, ICERM, Brown University – Providence, RI, USA Jan 2026 – Apr 2026
- Postdoctoral Researcher**, Tufts University – Somerville, MA, USA June 2025 – Present
- Working with [Elizabeth Newman](#) in structure-exploiting SciML and uncertainty quantification.
- Givens Intern**, Argonne National Laboratory – Lemont, IL, USA May 2024 – Aug 2024
- Givens Intern**, Argonne National Laboratory – Lemont, IL, USA May 2023 – Aug 2023
- Became a core maintainer of the PyOED project and published a peer reviewed paper in TOMS.
  - Published novel research in Robust Optimal Experimental Design, in review at SIAM CSE.
  - Under the mentorship under [Ahmed Attia](#).
- Research Assistant**, North Carolina State University – Raleigh, NC June 2021 – Present
- Published novel research in scalable UQ for infinite-dimensional Bayesian inverse problems.
  - Research Assistant for [Alen Alexanderian](#) via NSF Grant DMS-2111044.
- Teaching Assistant**, North Carolina State University – Raleigh, NC
- Led recitations for Calculus I, II, and the honors variants for 5 semesters.
- Undergraduate Research Assistant**, Ohio State University – Athens, OH May 2019 – Aug 2019
- Published novel research in the theory of pattern formation, for the 2D Swift-Hohenberg Equation.

## Publications

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- Boost Like a (Var)Pro: Trust-Region Gradient Boosting via Variable Projection** Mar 2026  
*Abhijit Chowdhary*, Elizabeth Newman, Deepanshu Verma  
<https://doi.org/10.48550/arXiv.2603.23658> (arXiv preprint)
- Scalable Uncertainty Quantification for Infinite-Dimensional Bayesian Inverse Problems** Feb 2025  
*Abhijit Chowdhary*  
[www.lib.ncsu.edu/resolver/1840.20/45106](http://www.lib.ncsu.edu/resolver/1840.20/45106) (PhD Thesis, NC State University)

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| <b>Robust optimal design of large-scale Bayesian nonlinear inverse problems</b><br><i>Abhijit Chowdhary</i> , Ahmed Attia, Alen Alexanderian<br><a href="https://doi.org/10.48550/arXiv.2409.09137">https://doi.org/10.48550/arXiv.2409.09137</a> (SIAM Journal on Scientific Computing)                                                                | Sept 2024 |
| <b>PyOED: An Extensible Suite for Data Assimilation and Model-Constrained Optimal Design of Experiments</b><br><i>Abhijit Chowdhary</i> , Shady E. Ahmed, Ahmed Attia<br><a href="https://dl.acm.org/doi/10.1145/3653071">https://dl.acm.org/doi/10.1145/3653071</a> (ACM Transactions on Mathematical Software)                                        | June 2024 |
| <b>Sensitivity Analysis of the Information Gain in Infinite-Dimensional Bayesian Linear Inverse Problems</b><br><i>Abhijit Chowdhary</i> , Shanyin Tong, Georg Stadler, Alen Alexanderian<br><a href="https://doi.org/10.1615/Int.J.UncertaintyQuantification.2024051416">https://doi.org/10.1615/Int.J.UncertaintyQuantification.2024051416</a> (IJUQ) | May 2024  |
| <b>Weak Diffusive Stability of Roll Solutions at the Zigzag Boundary</b><br><i>Abhijit Chowdhary</i> , Mason Haberle, William Ofori-atta, Qilian Wu<br><a href="https://doi.org/10.1137/24M1639683">https://doi.org/10.1137/24M1639683</a> (SIAM Journal of Mathematical Analysis)                                                                      | Oct 2023  |

## Software

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| <b>PyOED</b>                                                                                                                                                                                                                                                                                             | May 2023 – Present |
| <ul style="list-style-type: none"> <li>• Core maintainer of PyOED, an open source research software for model-constrained optimal design.</li> <li>• Designed to extend and integrate existing codes to rapidly test new developments.</li> <li>• Used in multiple research groups worldwide.</li> </ul> |                    |

## Talks

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| <b>VPBoost: Variable Projection for Gradient Boosting and VarProNets</b>                                                                                                                                                                                                                                                                        | Dec 2025 – Present  |
| <ul style="list-style-type: none"> <li>• CMO Integrating Data- and Physics-Driven Methods for Decision Making under Uncertainty, Oaxaca, MX (June 2026)</li> <li>• SIAM UQ 2026, Minneapolis, MN (Mar 2026).</li> <li>• ICERM, Providence, RI (Feb 2026).</li> <li>• CBMS Conference, University of Houston, Houston, TX (Dec 2025).</li> </ul> |                     |
| <b>Robust Optimal Experimental Design for Large-Scale Bayesian Inverse Problems</b>                                                                                                                                                                                                                                                             | May 2023 – Mar 2025 |
| <ul style="list-style-type: none"> <li>• SIAM CSE 2025, Fort Worth, TX (Mar 2025).</li> <li>• Applied Math Graduate Student Seminar, Raleigh, NC (Nov 2024).</li> <li>• SIAM MDS 2024, Atlanta, GA (Oct 2024).</li> <li>• SASSy Seminar, Argonne National Laboratory (May 2023).</li> </ul>                                                     |                     |
| <b>PyOED for Bayesian Optimal Experimental Design</b>                                                                                                                                                                                                                                                                                           | Feb 2024 – Aug 2024 |
| <ul style="list-style-type: none"> <li>• SASSy Seminar, Argonne National Laboratory (Aug 2024).</li> <li>• SIAM UQ 2024, Trieste, Italy (Feb 2024).</li> </ul>                                                                                                                                                                                  |                     |
| <b>Sensitivity Analysis and Scalable Optimal Design for Bayesian Inference</b>                                                                                                                                                                                                                                                                  | Apr 2022 – Aug 2023 |
| <ul style="list-style-type: none"> <li>• Applied Math Graduate Student Seminar, Raleigh, NC (Aug 2023).</li> <li>• SIAM CSE 2023, Amsterdam, Netherlands (Mar 2023).</li> <li>• Applied Math Graduate Student Seminar, Raleigh, NC (Apr and Sept 2022).</li> </ul>                                                                              |                     |

## Service

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| <b>Group Leader, VERSEIM-REU</b>                                                                                         | June 2026 – Aug 2026 |
| <ul style="list-style-type: none"> <li>• Leading a research group alongside <a href="#">Elizabeth Newman</a>.</li> </ul> |                      |

**Co-Organizer, Postdoctoral and Graduate Student Seminar, ICERM**

Jan 2026 – Apr 2026

**Lead Organizer, Applied Math Graduate Student Seminar (AMGSS), NCSU**

Jan 2022 – May 2025

**Peer-Review for the following journals.**

- Journal of Open Source Software (JOSS)
- SIAM Journal on Matrix Analysis and Applications (SIMAX)
- SIAM Journal on Uncertainty Quantification (JUQ)

## **Skills**

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**Programming:** Proficient in Python, Julia, C, C++, MATLAB, and Shell Scripting

**Technologies:** Generally familiar with the GPU programming interfaces, the machine learning stacks, Finite element software, high-performance linear algebra, and Linux.

**Mathematics:** Specialized in scientific computing, uncertainty quantification, and large-scale optimization. Theoretical foundations in the PDE, probability, and functional analysis.

**Languages:** English (native), Latin, Hindi